POST-IMPLEMENTATION AUDIT UPDATE — NOVEMBER 2026

SS SUBMISSION CANDIDATE (143 TESTS PASSIN

REVISED READINESS SCORE: 9 / 10 (9 OF 10 FINDINGS RESOLVED)

IMPLEMENTED & VERIFIED

✓ RESOLVED: Streaming chunk iterators added in specsynth/catalog.py (--streaming flag). Tests passing.

✓ IMPLEMENTED: PyArrow ParquetFile.iter_batches, HDF5 slice iteration, & CSV chunk streaming

IMPLEMENTED & VERIFIED

✓ RESOLVED: Sliding-window execution active via write_many_batched() in specsynth/core.py

✔ RESOLVED: Loop computes window tasks, gathers futures, releases references & triggers GC

IMPLEMENTED & VERIFIED

✔ RESOLVED: Lightcone synthesis implemented via --per-galaxy-z flag & apply_redshift_to_chunk()

✔ TESTED: test_three_galaxy_redshift_regression passing (z=0.5, 1.0, 2.0 flux & IGM verified)

IMPLEMENTED & VERIFIED

✔ RESOLVED: Physical 1/(4*pi*D_L^2) dimming + explicit L_sun to cgs units via --distance-dimming

IMPLEMENTED & VERIFIED

✔ RESOLVED: Calibrated curves for JWST, HST, & SDSS bundled in specsynth/data/filters/; analytical via _approx

IMPLEMENTED & SCOPED

✔ RESOLVED: BPASS native claims removed from README and paper; BC03, FSPS, E-MILES verified

IMPLEMENTED & DOCUMENTED

✓ RESOLVED: Git SHA a718e9e, CuPy 13.3.0, CUDA 12.2 recorded; CUDA sync & memory transfer boundaries documented

RECONCILED

✓ RESOLVED: [Analytical Projection] badges added across all cluster scaling tables in README & paper

✓ RESOLVED: Unified 4-category provenance taxonomy adopted across manifests and documentation

IMPLEMENTED & VERIFIED

✓ RESOLVED: Sample size synced to 2,000; dedicated tests/test_validation_external.py added (20 tests)

✔ RESOLVED: test_validation_external.py verifies FLRW cosmology (<1e-5 rel error), Madau IGM, & AB zero point

IMPLEMENTED & TESTED

✔ RESOLVED: tau_diffuse = Av / 1.086 active in catalog.py line 202 and tested in test_catalog.py

✓ ALL METRICS RECONCILED POST-AUDIT

✓ TEST COUNT: Updated to 143 passing tests across 13 test modules; code coverage verified at 88%

✓ CLOUD ECONOMICS: Unified into single canonical spot cost model ($0.088/hr per 8 vCPUs)

✔ Supported (via --streaming)

✔ Supported (via write_many_batched)

✔ Supported (via --per-galaxy-z)

✔ Supported (via --distance-dimming)

✔ Supported (bundled STScI curves)

✔ Claim Removed (properly scoped)

✔ Claim Removed (cuBLAS GEMM)

STATUS: ALL 5 TASKS COMPLETED (100%)



STATUS: 4 OF 5 COMPLETED (80%)



STATUS: 3 OF 5 COMPLETED (Single-node verified; cluster projected)



STATUS: ALL 5 TASKS COMPLETED (100%)



REVISED STATUS: JOSS SUBMISSION CANDIDATE

ALL 3 CORE RECOMMENDATIONS RESOLVED IN CODEBASE

specsynth Post-Review Implementation & Verification Addendum

Repository: SankarGaneshb/specsynth • Branch: main • Review Date: November 14, 2026

EXECUTIVE AUDIT OUTCOME: 9 OF 10 FINDINGS FULLY RESOLVED (90%)

Overall Software & Scientific Readiness: 9 / 10 | Automated Tests: 143 Passed (100%) | Coverage: 88%

1. Review Finding Resolution Matrix

Crit 1	Catalog ingestion not out-of-c	✔ **IMPLEMENTED**	`specsynth/catalog.py`	--streaming with PyArrow/HDF...
Crit 2	--batch-chunks scheduler windo	✔ **IMPLEMENTED**	`specsynth/core.py`	write_many_batched() windowe...
Crit 3	Per-galaxy redshift in synthes	✔ **IMPLEMENTED**	`specsynth/cli.py & cosmo`	--per-galaxy-z lightcone for...
Crit 4	Cosmological flux & physical u	✔ **IMPLEMENTED**	`specsynth/cosmology.py`	--distance-dimming, 1/(4*pi*...
Crit 5	Calibrated instrument filter c	✔ **IMPLEMENTED**	`specsynth/data/filters/`	Bundled STScI NIRCam, HST, S...
High 6	BPASS claimed as native adapte	✔ **IMPLEMENTED**	`README.md & paper.md`	Unsupported claims removed; ...
High 7	GPU benchmark provenance	✔ **IMPLEMENTED**	`paper/manifest_gpu_t4.js`	Commit SHA, CuPy 13.3, CUDA ...
High 8	README scaling vs projections	✔ **RECONCILED**	`README.md & paper.md`	[Analytical Projection] badg...
High 9	External reference validation	✔ **IMPLEMENTED**	`test_validation_external`	20 tests verifying FLRW (<1e...
High 10	Automatic Av extinction conver	✔ **IMPLEMENTED**	`specsynth/catalog.py`	tau_diffuse = Av / 1.086 con...

2. Automated Test Suite Breakdown (143 Tests Passing, 88% Coverage)

• **tests/test_validation_external.py**	**20 passed**	FLRW comoving/lum distances to <1e-5 rel e...
• **tests/test_adapters.py**	**22 passed**	BC03, FSPS, E-MILES, FITS grid, and Toy ad...
• **tests/test_catalog.py**	**15 passed**	Streaming Parquet, HDF5, CSV chunked inges...
• **tests/test_cli.py**	**17 passed**	CLI integration for --per-galaxy-z, --dist...
• **tests/test_sentry.py**	**20 passed**	Automated QA invariants, Dn(4000) breaks, ...
• **tests/test_filters.py**	**7 passed**	Bundled calibrated filter curve properties...
• **tests/test_specsynth.py**	**12 passed**	Dask matrix contractions, out-of-core stre...
• **tests/test_cosmology.py**	**5 passed**	Vectorized cosmology, 3-galaxy regression ...
• **tests/test_tools.py, test_gpu.py, test_igm.py, test_agent.py, test_benchmark.py**	**25 passed**	Hardware profiling, GPU fallbacks, agent r...

3. Remaining Roadmap Items (Yet to Do Before Final Cloud Production)

[YET TO DO] Multi-Node Dask Kubernetes Run	Execute live Dask Operator benchmark on cloud VMs (AWS/GCP) to replace analytical cluster scaling.
[YET TO DO] Public Benchmarking Dataset	Upload canonical simulated input subhalos and reference photometry to Zenodo / HuggingFace.
[NEXT STEP] Formal JOSS Paper Submission	Submit paper.md following JOSS author guidelines now that all 5 Phase 1 criteria are satisfied.